Friction: Test the Surfaces · Teacher and Cover Plan

Learning goal: I can compare friction on different run-out surfaces using repeated measurements and a controlled setup.

Scheme Outcome

TRACK Science scheme of work: Forces and Friction

Compare run-out surfaces using controlled release and repeated stopping-distance measurements in centimetres.

Success Criteria

I can keep the block, ramp height, and release point the same.

I can record repeated stopping distances in centimetres.

I can use the evidence to identify which surface gave greater resistance.

Model Sequence (I do)

Keep the same block, low ramp, release mark, and no-push release. Change only the run-out surface.

Measure from the same zero mark at the bottom of the ramp to the same front edge of the block in cm.

Repeat each surface twice and keep both results·shorter stopping distance shows greater resistance.

Look at both trials before claiming a winner: Surface A (46 cm, 48 cm) vs Surface B (24 cm, 26 cm).

Surface B stopped the block sooner in both trials, supporting greater friction on B in this setup.

If a trial has a known release mistake, keep the number in your table, note the mistake, and run a retry.

We Do & Hinge Check

Which method step gives a valid comparison of two run-out surfaces?

Answer: A. Keeping the block, ramp height, and release identical isolates the effect of the run-out surface.

Safe Practical & Card Setup

Compare two run-out surfaces twice using the low ramp setup (or analyse the supplied reference table) and write a conclusion.

Kit: Low secured ramp, light sliding block, two run-out surfaces, metre rule/tape, padded stop, or supplied reference data table.

Keep the ramp low on the table or floor tray with a padded stop at the end. Keep fingers clear of the block path.

You Do Routes (Supported, Independent, Aspire)

Supported: Compare Surface A (46 cm, 48 cm) and Surface B (24 cm, 26 cm). Circle the shorter distance each time and name the grippier surface. (Help: 24 cm is shorter than 46 cm. Surface __ stopped the block

sooner, so it had more __.)

Independent: Record two trials for Surface A and Surface B (practical or supplied data: A = 46, 48 cm; B = 24, 26 cm). Write a conclusion naming the numbers and one control. (Help: Surface __ stopped the block in __ cm and __ cm, compared with __ cm and __ cm on __. This shows greater friction on __ because __.)

Aspire: Explain why we repeat each surface and state one limitation of our two-surface comparison. (Help: Repeating checks consistency. Our results apply to this __ and release speed, not every possible surface.)